About the course

Human Exploration of the Physical Universe

- Plato, Lindberg
- Aristotle, Lindberg
- Cohen, Newton

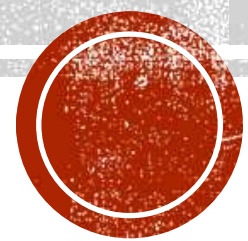
Human Exploration of the World of Life

- Darwin
- Watson
- Carson

Our Understanding of Human Understanding

- Poincare
- Kandel
- Needham
- Sivin, Shen Kuo
- Dunhan, Euclid

GFN1000
IN DIALOGUE WITH NATURE
与自然对话



Jules Henri Poincaré

Science and method

CORE QUESTION

What guides scientists?

Methods & Motivations



Content



Text 7: Henri Poincaré (詹昂利·庞加莱)

"Science and Method" (科学与方法):

Chapter I *The Selection of Facts (事实的选择)*

Chapter III *Mathematical Discovery (数学创造)*

Henri Poincaré

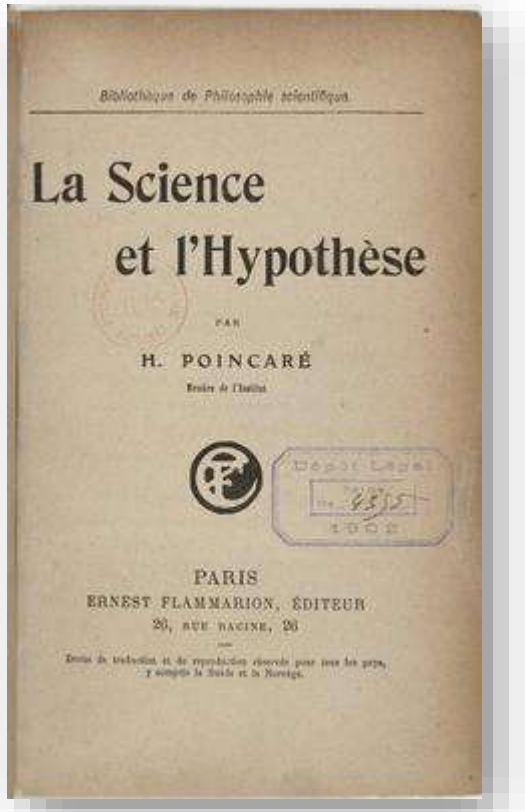


(1854-1912)

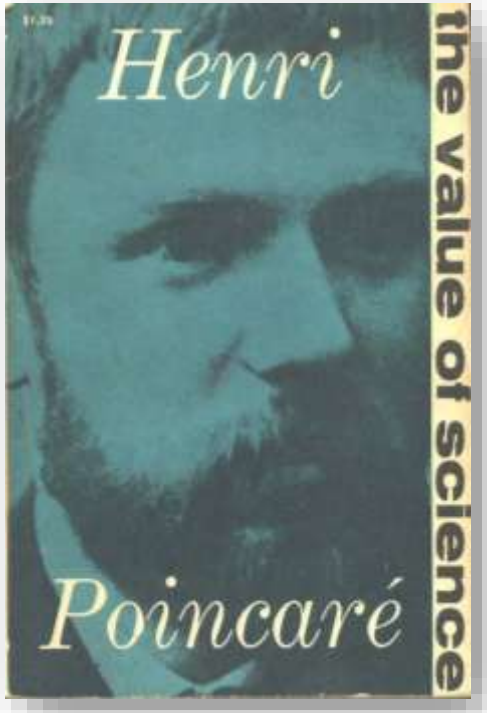
- A French mathematician, theoretical physicist, engineer, and philosopher of science, considered as the last polymath (博学家).
- The only member elected to every one of the five sections of the French Academy of Sciences: geometry, mechanics, physics, geography, and navigation sections.
  - RAS (Royal Astronomical Society) Gold Medal (1900)
  - Matteucci Medal (1905): an Italian award for his contributions in physics.
  - Bolyai Prize (1905): international prize founded by the Hungarian Academy of Sciences to award mathematicians for monographs with important new results in the preceding 10 years.
  - Bruce Medal (1911): for outstanding lifetime contributions to astronomy.

Henri Poincaré (1854-1912)

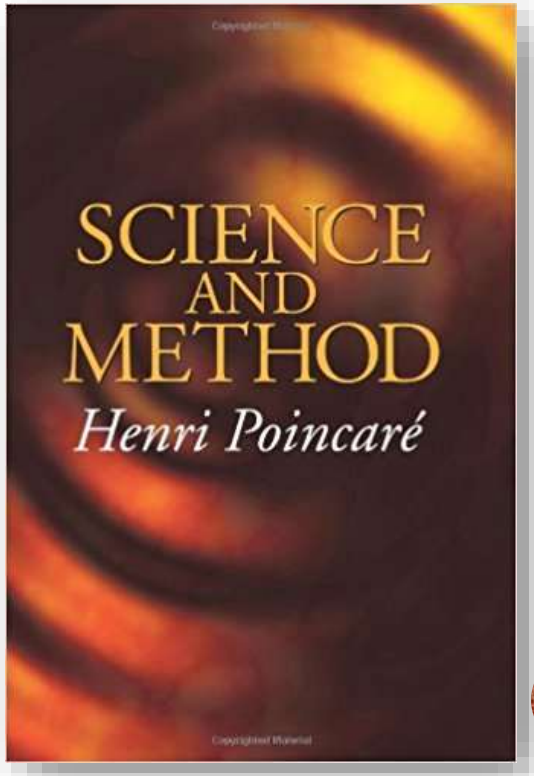
- *Science and Hypothesis (1902)*



- *The Value of Science (1905)*



- *Science and Method (1908)*



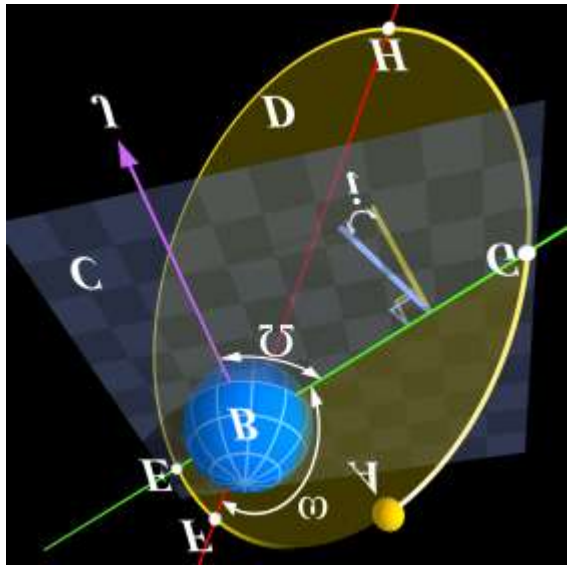
Solvay Conference (1911, Brussels)



N-body problem

- A problem of predicting the individual motions of a group of celestial objects interacting with each other gravitationally (n>2).

- The problem of finding the general solution to the motion of more than two orbiting bodies in the Solar System had eluded mathematicians since Newton's time.



King Oscar II of Sweden (1829-1907)



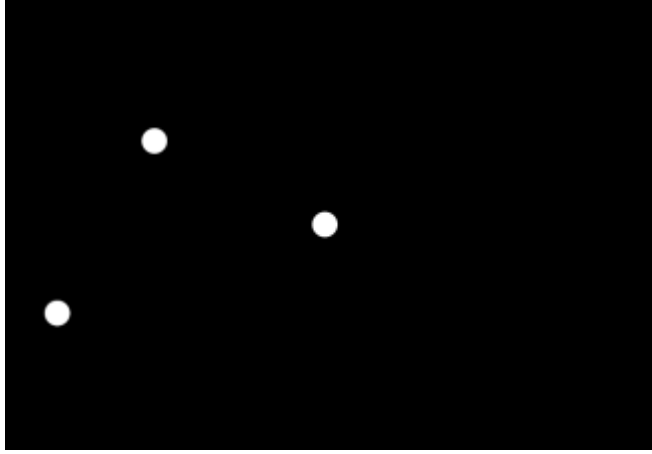
- A generous sponsor of the sciences, particularly mathematics.

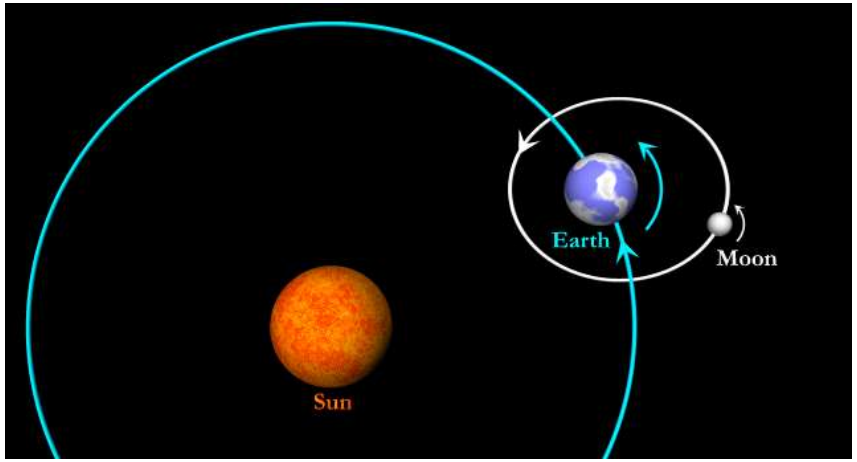
- In 1885, a math contest was announced as a celebration of the 60th birthday of King Oscar II of Sweden and Norway. One of the problems was:

  "Given a system of arbitrarily many mass points that attract each according to Newton's law, under the assumption that no two points ever collide, try to find a representation of the coordinates of each point as a series in a variable that is some known function of time and for all of whose values the series converges uniformly."

- A contest for any scientist who was able to answer the question of the n-body problem, specifically applied to the stability of the Solar System.

Three-body problem



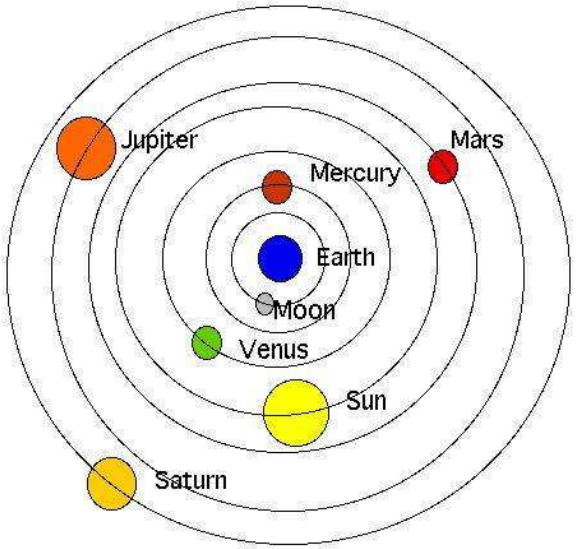


- Poincaré focused on the 3-body problem
  - The three-body problem is a special case of the n-body problem.
- Poincaré won the prize.
  - The original version of his manuscript contained a serious error.
  - The version finally printed contained many important ideas which led to the development of chaos theory.
- The problem as stated originally was finally solved by Karl Fritiof Sundman for n = 3 in 1907.

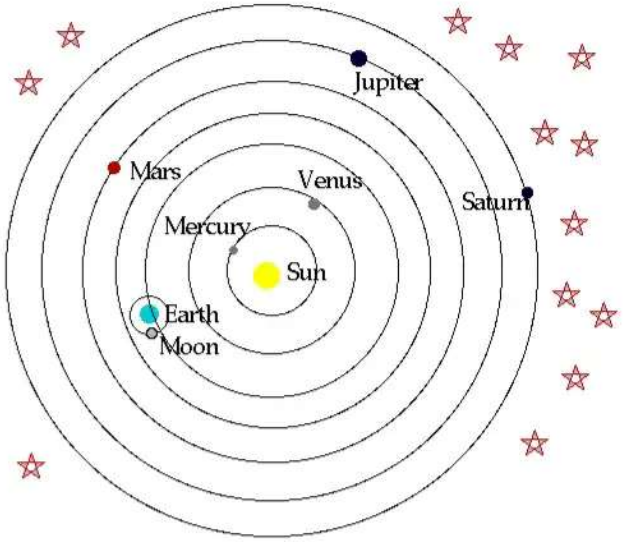
Poincaré found that there could be chaos in 3-body motion under mutual gravity!

Stability

Plato (428 – 347 BC): The most perfect form is the circle → objects in heaven move in circles with uniform motion.



Aristotle's Geocentric model



Copernicus's Heliocentric model

Newton (1642-1727): mechanical, stable, regular, predictable motion

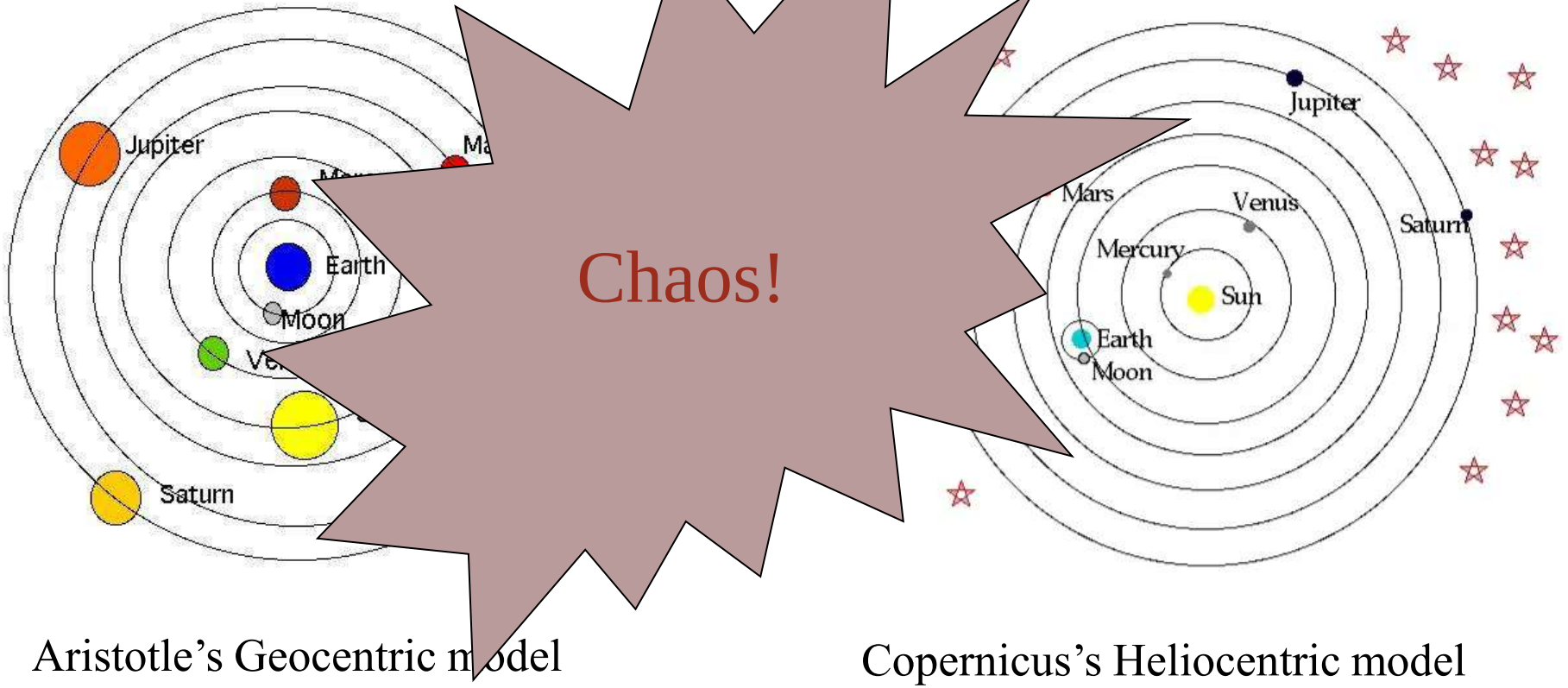
Newtonian Mechanics Has Dominated for 200 Years Since the Late 17th Century.

- Newtonian mechanics was considered encompassing everything in science.
- *"There is nothing new to be discovered in physics now. All that remains is more and more precise measurement."*
- Physics is not promising anymore! Go for social sciences.
- However, unexpected phenomena discovered.



Stability

Plato (428 – 347 BC): The most perfect form is the circle →objects in heaven move in circles with uniform motion



Aristotle's Geocentric model

Copernicus's Heliocentric model

Newton (1642-1727): clockwork, mechanical, stable, regular, predictable motion

Chaos theory (混沌理论)

- Chaos theory:
  focused on underlying patterns and deterministic laws highly sensitive to initial conditions in dynamical systems that were thought to have completely random states of disorder and irregularities.

- E.g. double pendulum (双摆)

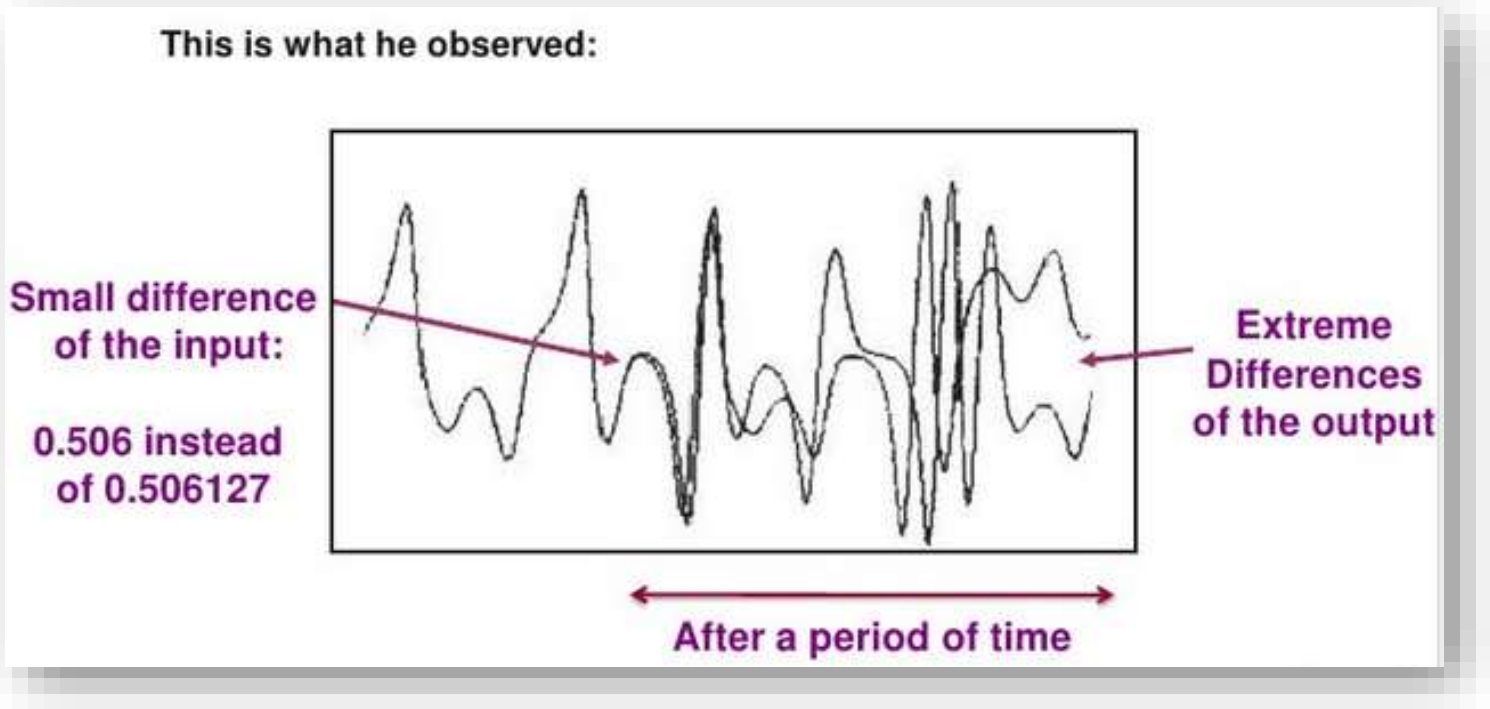


Chaos theory

The Butterfly Effect (蝴蝶效应)

- One flap of a butterfly's wings could change the course of weather forever!
- Some model weather systems are very sensitive to initial conditions.

  Edward Lorenz (1961): changing a parameter from 0.506127 to 0.506 yields a completely different weather scenario! (The limitation of weather prediction)

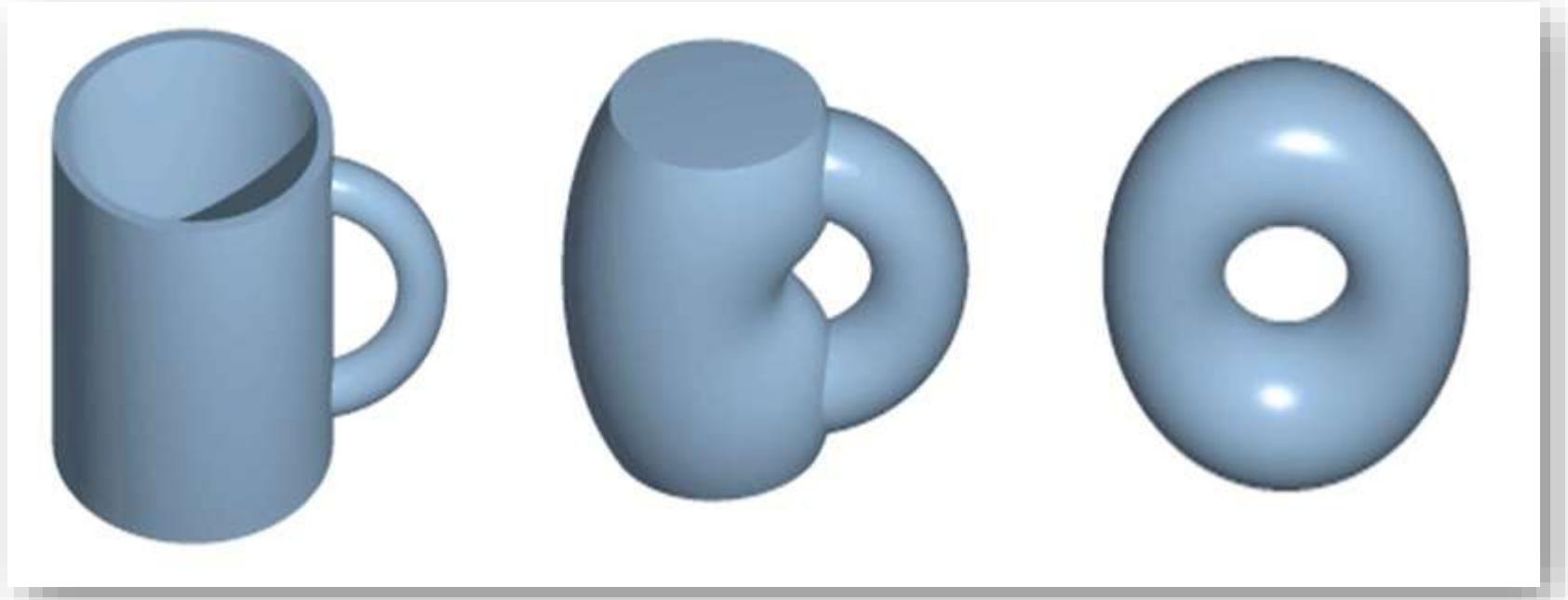


Poincaré conjecture (庞加莱猜想)

-solved in 2002-2003 by Grigori Perelman

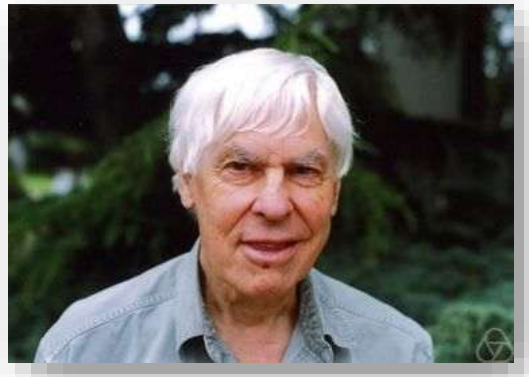
Every simply connected, closed 3-manifold is homeomorphic to the 3-sphere.

任一单连通的、封闭的三维流形与三维球面同胚。



Poincaré conjecture (庞加莱猜想)

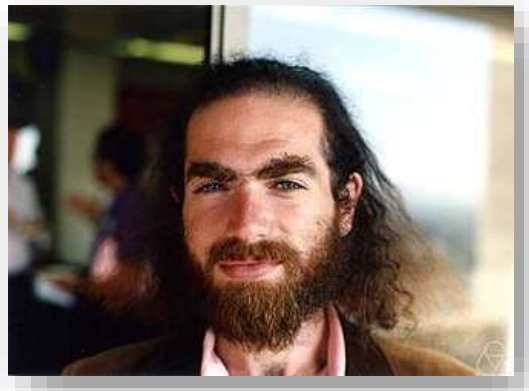
- Steven Smale (斯蒂芬·斯梅尔)(1930-)
  - proved all dimensions greater than or equal to five in 1961
  - awarded the Fields Medal in 1966



- Michael Freedman (迈克尔·弗里德曼)(1951-)
  - proved in four dimension in 1982
  - awarded the Fields Medal in 1986



- Grigori Perelman (格里戈里·佩雷尔曼)(1966-)
  - sketched a proof of the conjecture and a more general conjecture, which is Thurston's geometrization conjecture in 2002
  - awarded the Fields Medal in 2006, declined



After Perelman…

Over the next several years, several mathematicians studied Perelman's papers and produced detailed formulations of his work.

- Bruce Kleiner(克莱纳) and John W. Lott(洛特)
  - filled in the details of Perelman's proof of the geometrization conjecture




- Huai-Dong Cao(曹怀东) and Xi-Ping Zhu(朱熹平)
  - completed proof of the Poincaré and geometrization conjectures




- John Morgan (摩根) and Gang Tian (田刚)
  - gave a detailed proof of just the Poincaré Conjecture and expanded this to a book

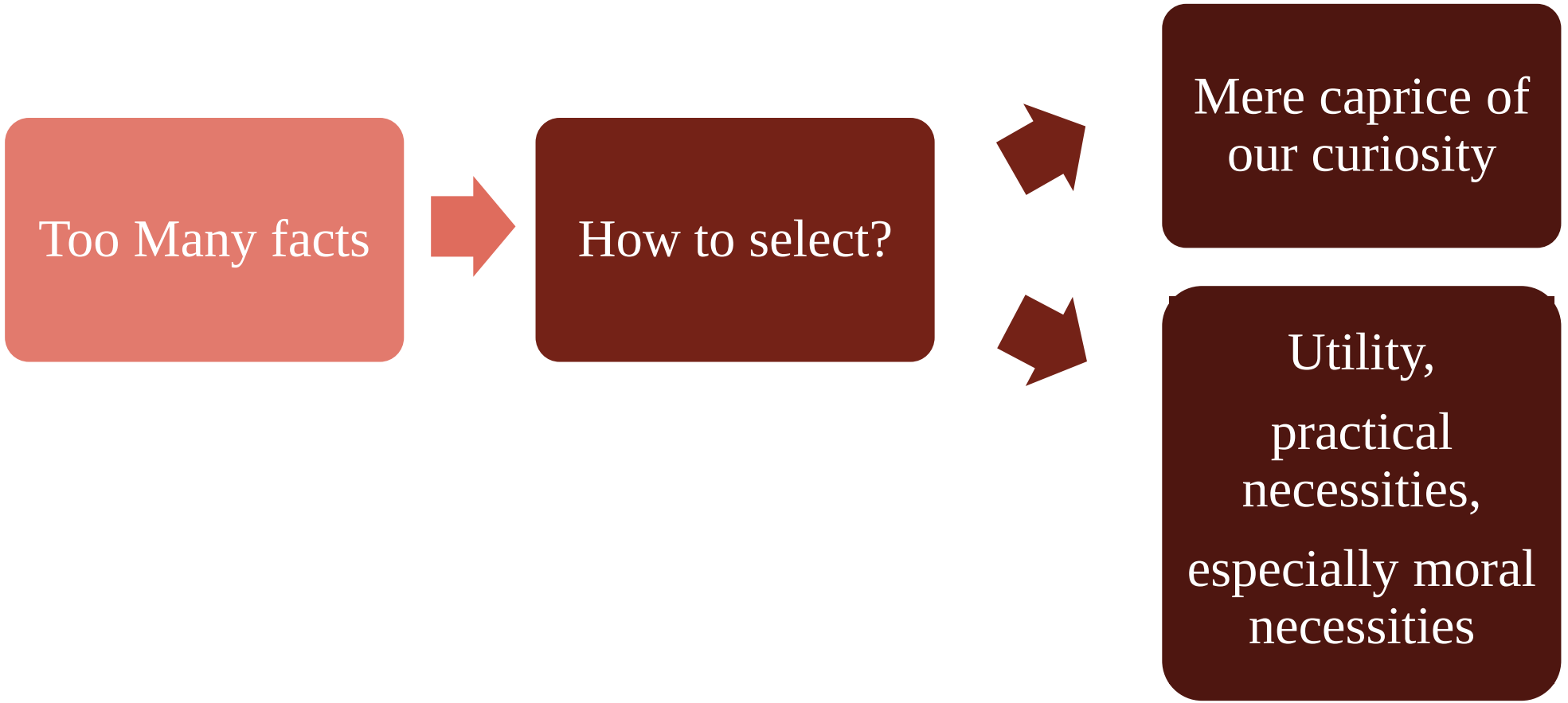



Chapter I

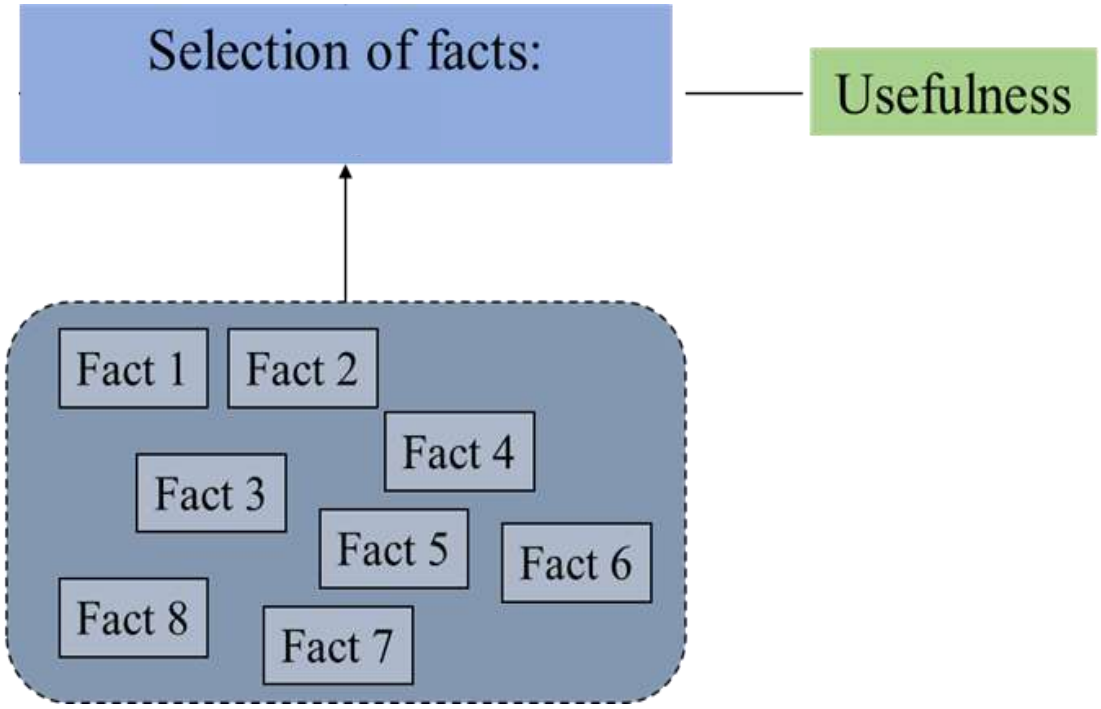
The selection of facts



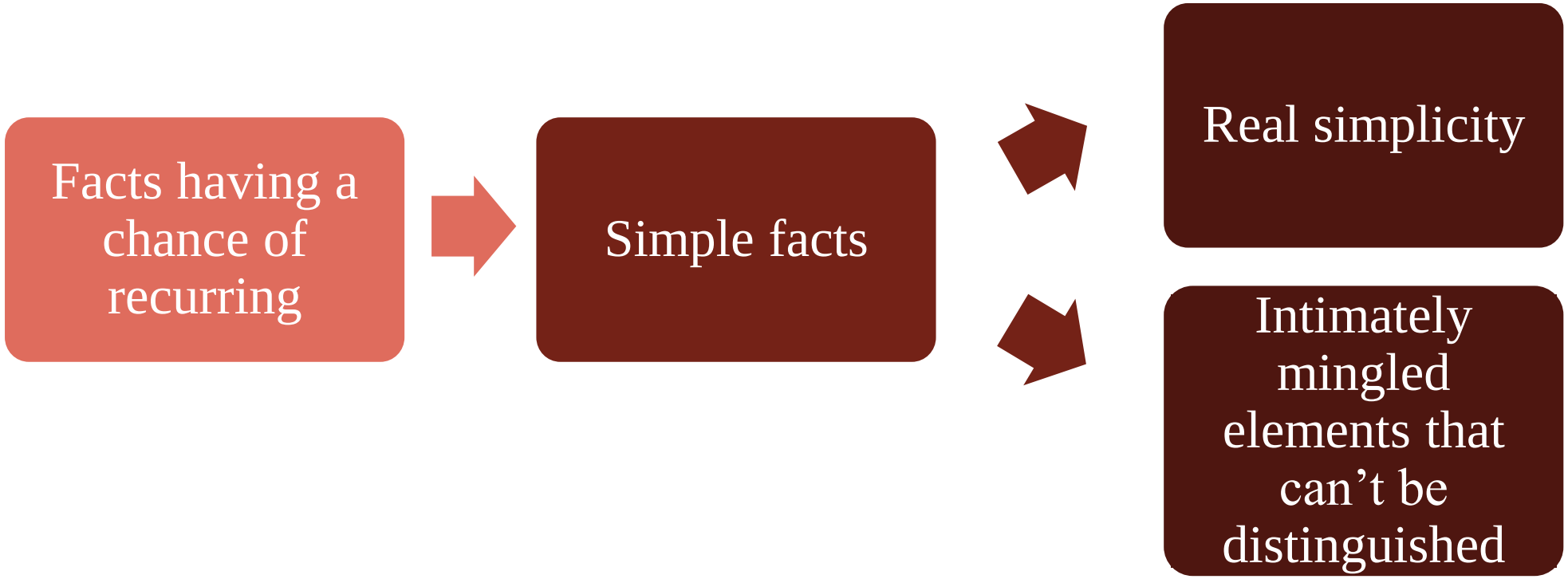
How do we select what to study?



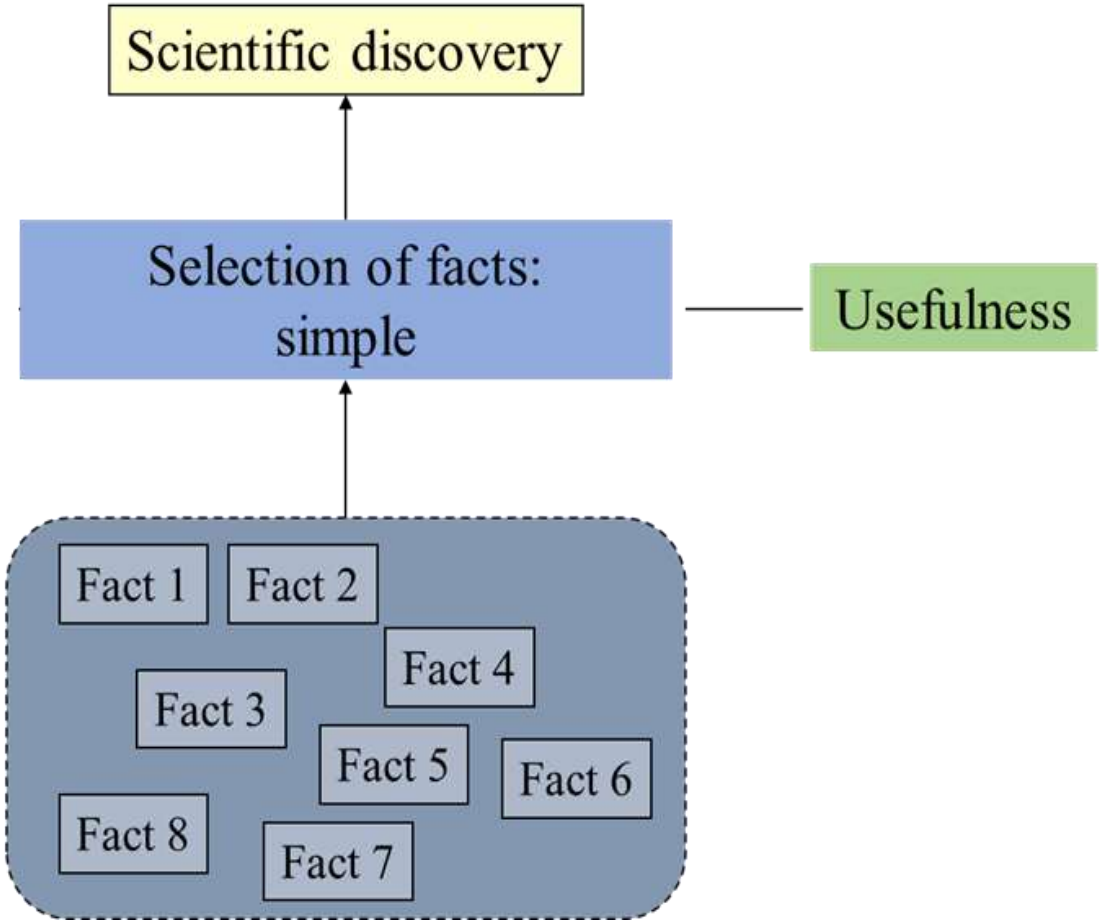
How do we select what to study?



How do we select what to study?



How do we select what to study?



Where to find simple facts?

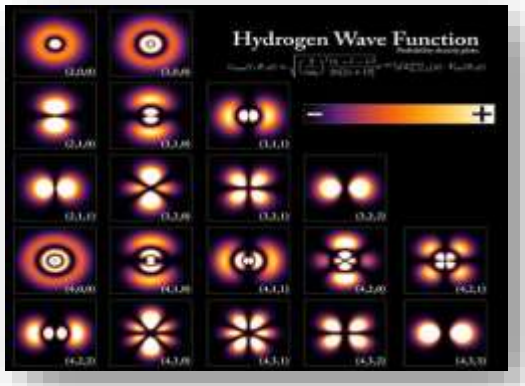
Where to find simple facts?

Infinitely great

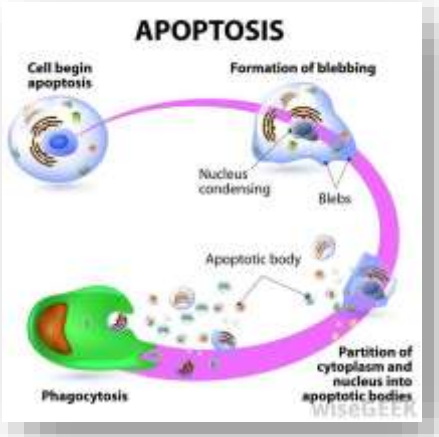
- *Astronomer*



Infinitely small



- *Physicist*



- *Biologist*

Where to find simple facts?

Why the scientist studies what he studies?

But so far we have only considered one side of the question. The scientist 15
does not study nature because it is useful to do so. He studies it because he takes pleasure in it, and he takes pleasure in it because it is beautiful. If nature were not beautiful it would not be worth knowing, and life would not be worth living. I am

- He studies it because he takes pleasure in it, and he takes pleasure in it because it is **beautiful**.
- Not sensual beauty, but **intellectual beauty.**
- Intellectual beauty comes from the **harmonious order** of its parts.

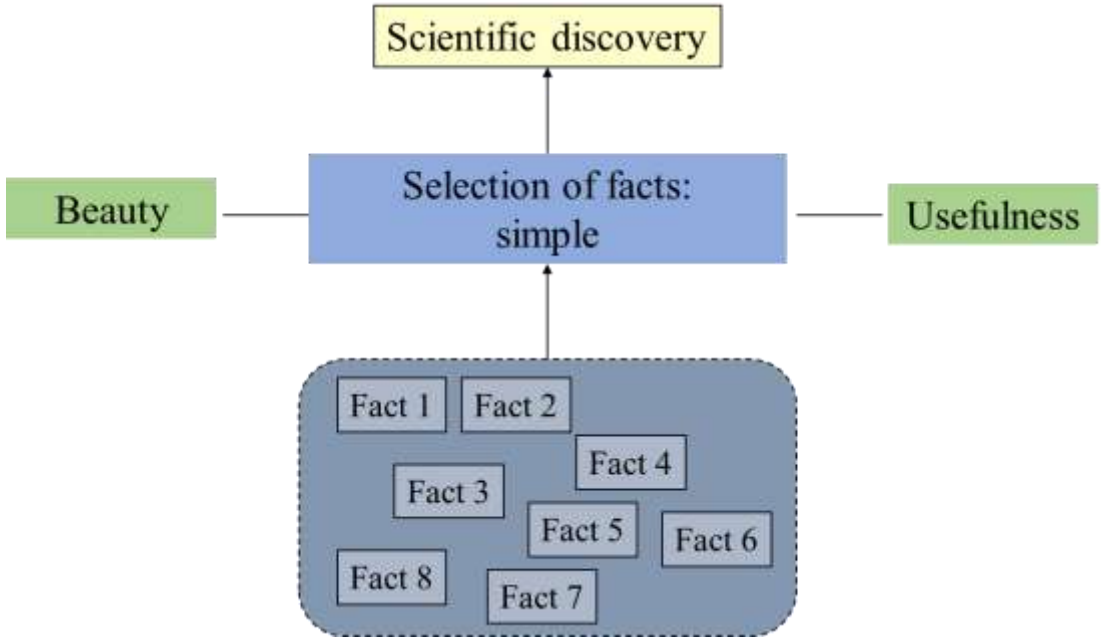
The Driving Force/Selection Criteria

16 It is, then, the search for this special beauty, the sense of the harmony of the world, that makes us select the facts best suited to contribute to this harmony; just as the artist selects those features of his sitter which complete the portrait and give it character and life. And there is no fear that this instinctive and unacknowledged

- a sense of beauty (Para. 15)
- the beauty = the **harmony**, facts contributing to the harmony are chosen. (Para. 16)

17 It is because simplicity and vastness are both beautiful that we seek by preference simple facts and vast facts; that we take delight, now in following the giant courses of the stars, now in scrutinizing with a microscope that prodigious smallness which is also a vastness, and now in seeking in geological ages the traces of a past that attracts us because of its remoteness.

The Flow



18 Thus we see that care for the beautiful leads us to the same selection as care for the useful. Similarly economy of thought, that economy of effort which, according to Mach, is the constant tendency of science, is a source of beauty as well as a practical advantage. The buildings we admire are those in which the architect has succeeded in proportioning the means to the end, in which the columns seem to carry the burdens imposed on them lightly and without effort, like the graceful caryatids of the Erechtheum.

Beauty of mathematics

- What kinds of intellectual beauty that mathematics has?

S H O C S

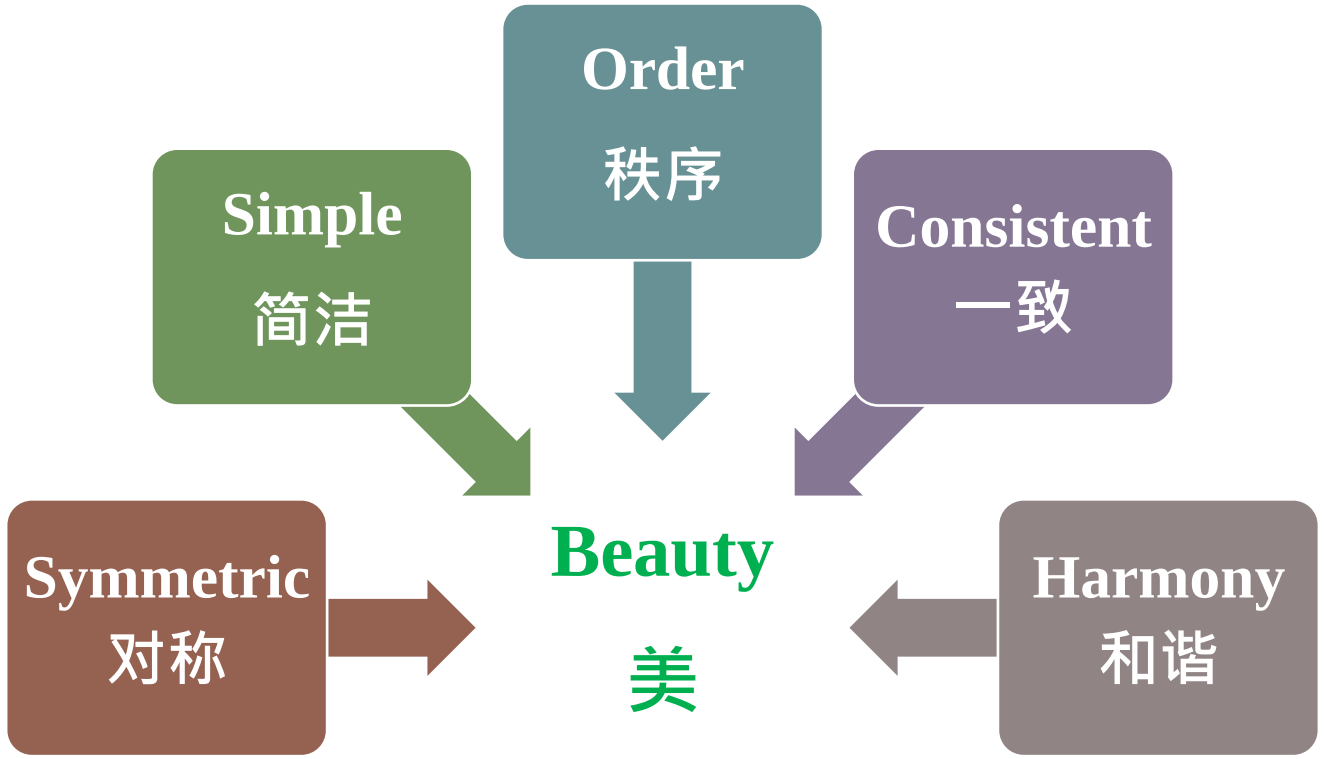
Simple 简洁

Harmonious 和谐

Order 秩序

Consistent 一致

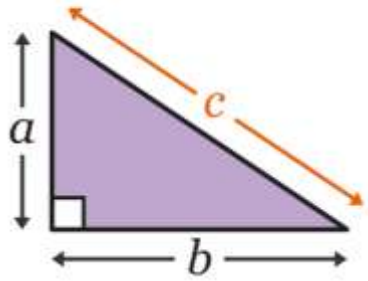
Symmetric 对称



What is beautiful to you ?







$$a^2 + b^2 = c^2$$

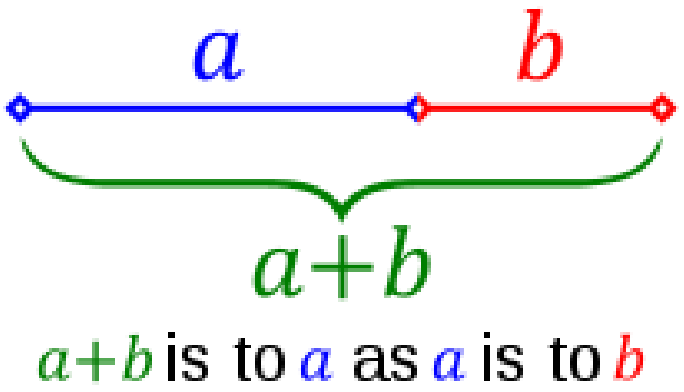
蒹葭苍苍，白露为霜。
所谓伊人，在水一方。
溯洄从之，道阻且长。
溯游从之，宛在水中央。

Do you think mathematics is as beautiful as poetry, painting, architecture…

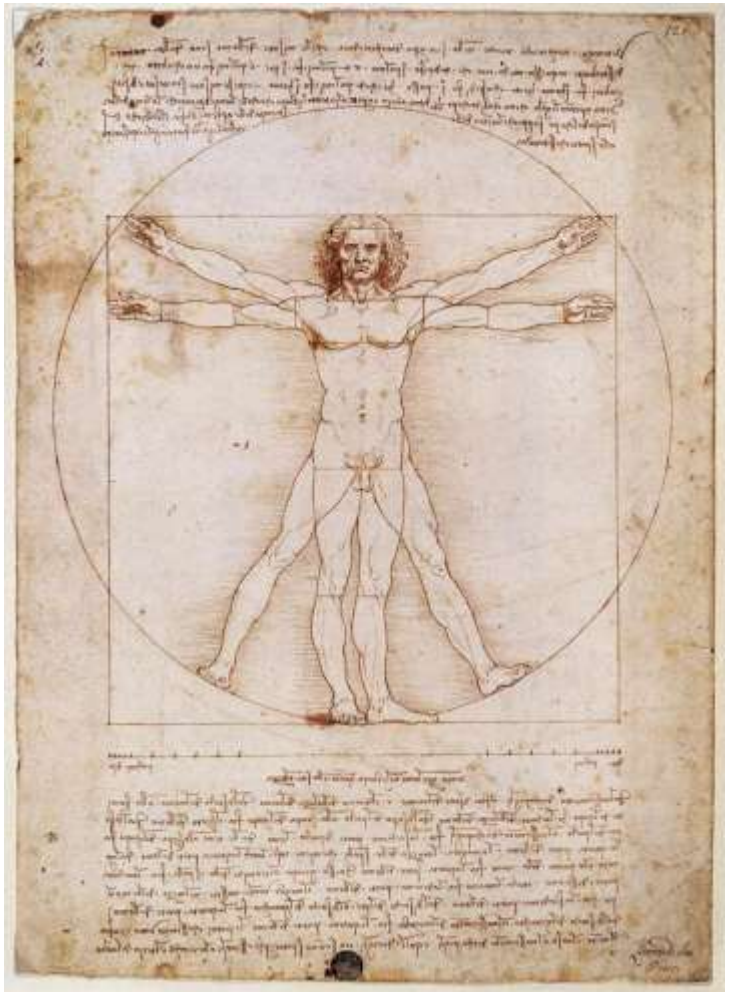
Example of beautiful math

The Golden Ratio (黄金分割)

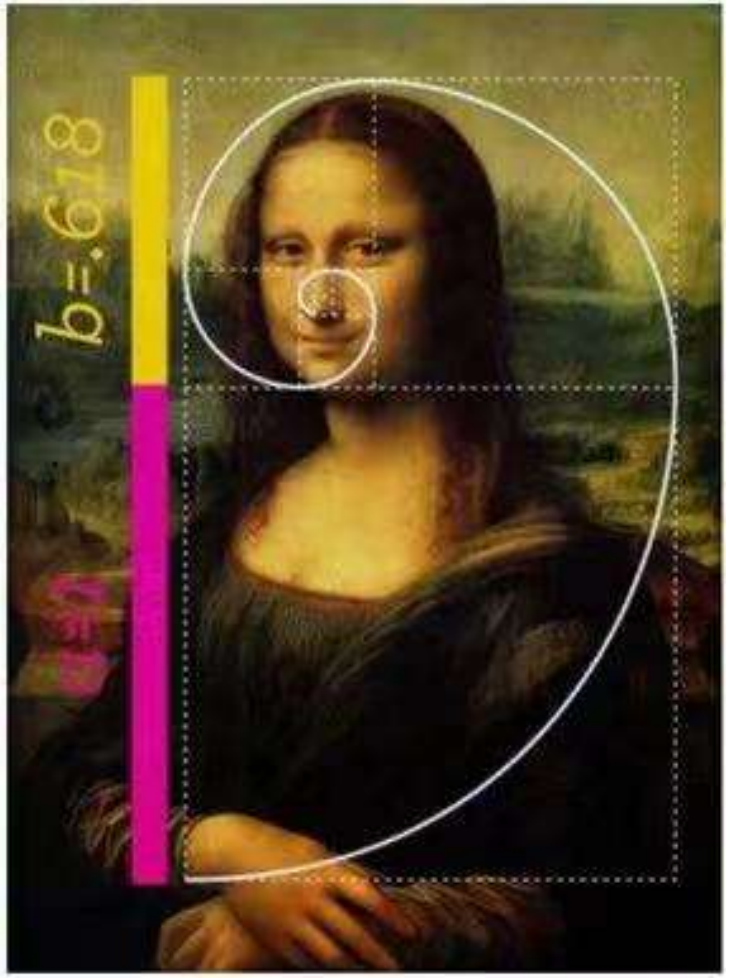
- One of most well known numbers related to beauty.
- First explicitly defined in Euclid's "*Elements*" with the name DEMR ("Division in Extreme and Mean Ratio").
- Also called golden mean, golden section, golden proportion, golden cut, golden number, …
- Translated in Chinese as 黄金比例, 黄金率, 黄金分割, …



Leonardo Da Vinci



Vitruvian Man



Mona Lisa

Chapter III

Mathematical discovery



Why did Poincaré choose mathematical discovery?

The genesis of mathematical discovery is a problem which must inspire the psychologist with the keenest interest. For this is the process in which the human mind seems to borrow least from the exterior world, in which it acts, or appears to act, only by itself and on itself, so that by studying the process of geometric thought we may hope to arrive at what is most essential in the human mind.

(Paragraph 1)



What is important for being a mathematician?

---Memory, Attention and Intuition

it in a mechanical way, and then, if his memory fails him, he may apply it wrongly. It is thus, to take a simple and almost vulgar example, that we sometimes make mistakes in calculation, because we have forgotten our multiplication table.

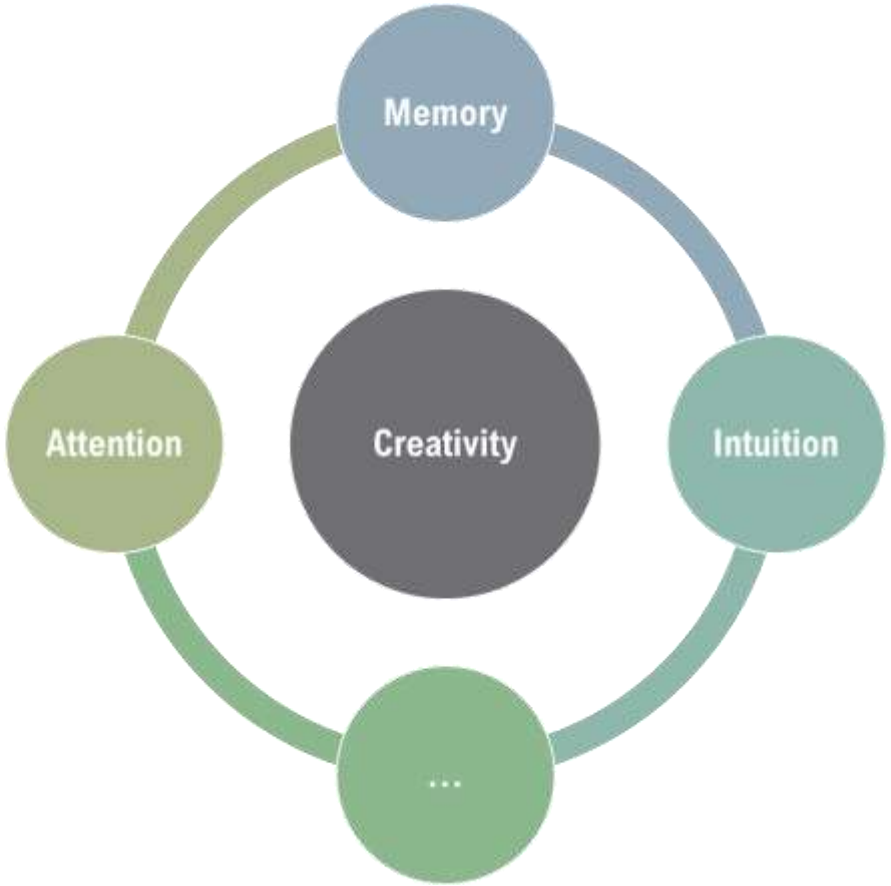
8 On this view special aptitude for mathematics would be due to nothing but a very certain memory or a tremendous power of attention. It would be a quality

It seems to me, then, as I repeat an argument I have learnt, that I could have 11
discovered it. This is often only an illusion; but even then, even if I am not clever enough to create for myself, I rediscover it myself as I repeat it.

We can understand that this feeling, this intuition of mathematical order, 12
which enables us to guess hidden harmonies and relations, cannot belong to everyone. Some have neither this delicate feeling that is difficult to define, nor a power of memory and attention above the common, and so they are absolutely

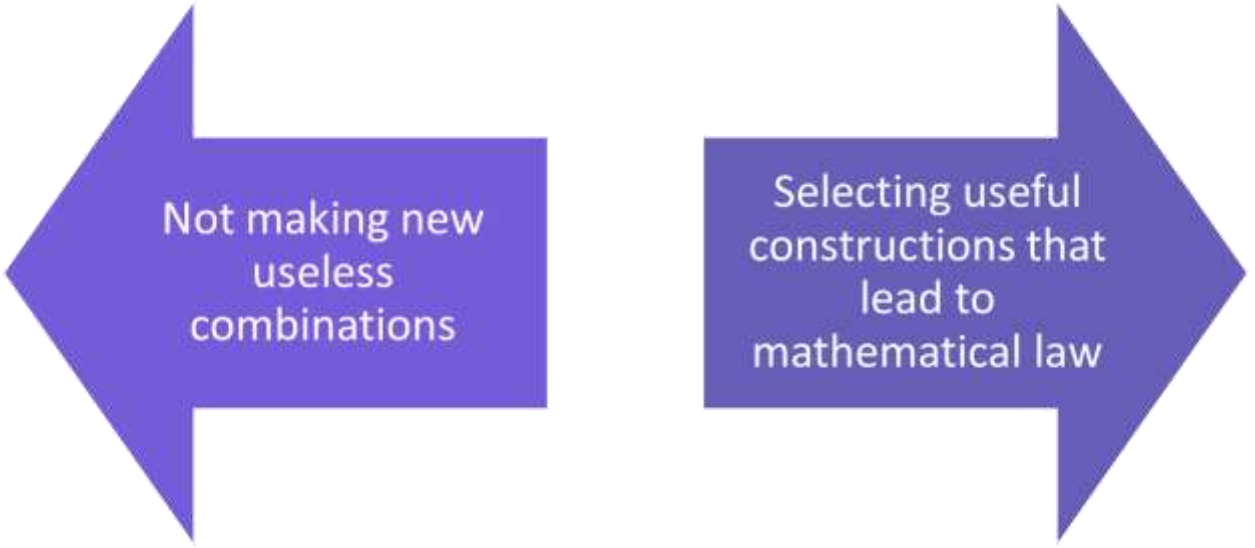
“It is by logic that we prove, but by intuition that we discover. To know how to criticize is good, to know how to create is better.”

Science and Method

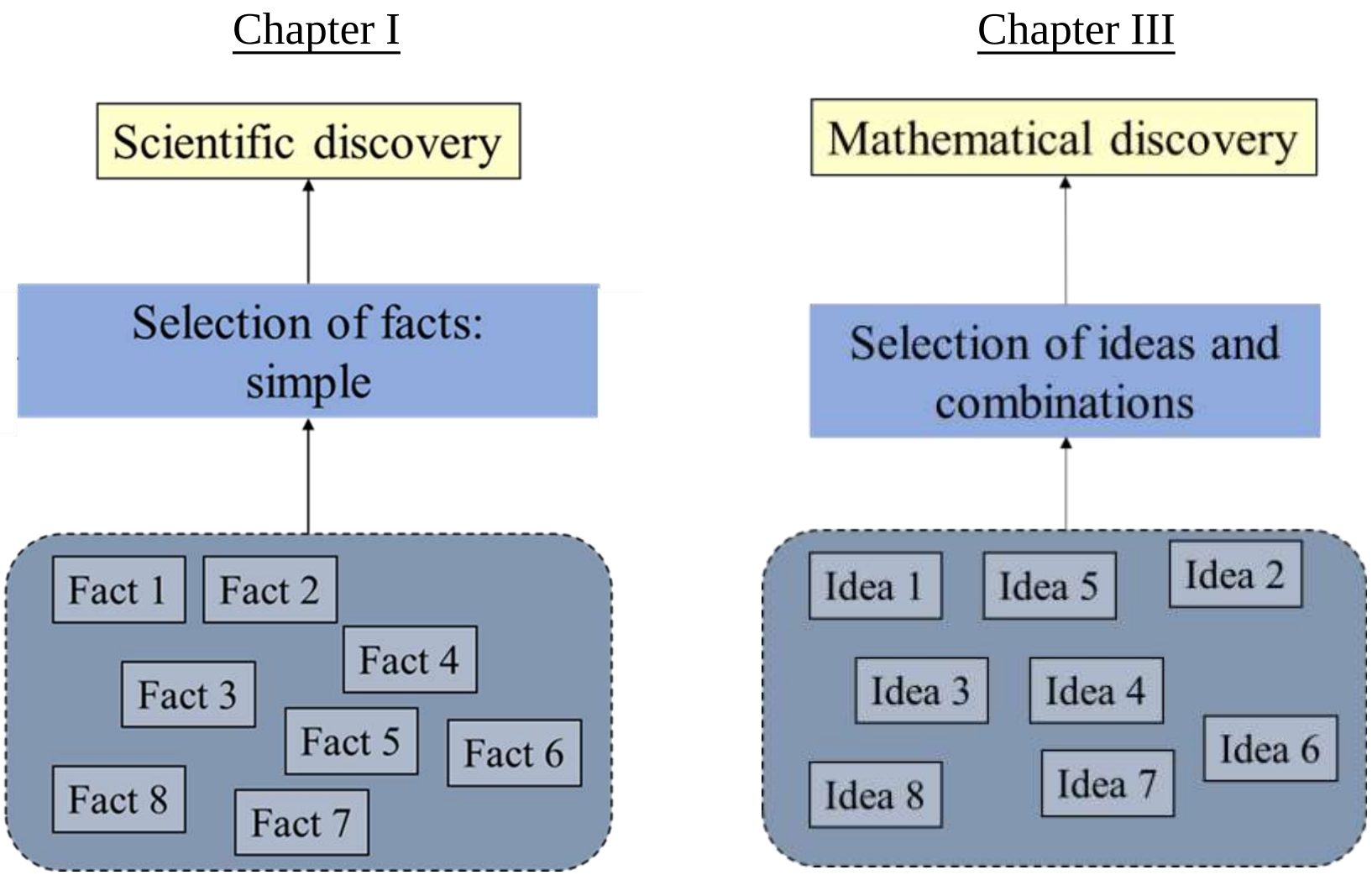


What is mathematical discovery?

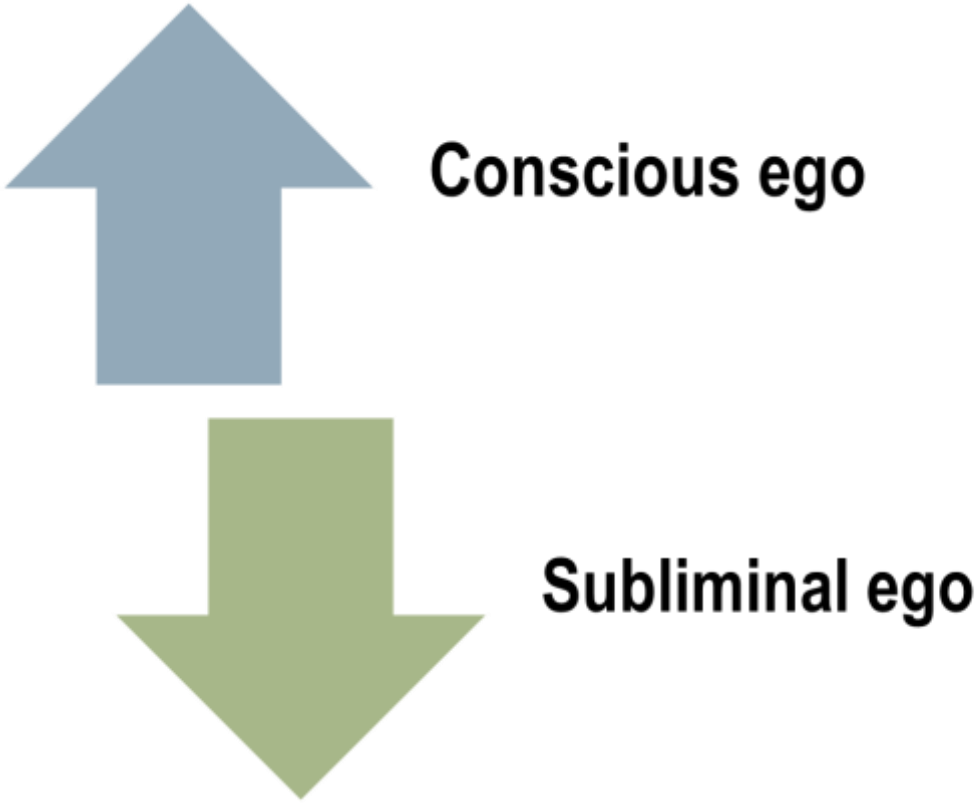
13 What, in fact, is mathematical discovery? It does not consist in making new combinations with mathematical entities that are already known. That can be done by any one, and the combinations that could be so formed would be infinite in number, and the greater part of them would be absolutely devoid of interest. Discovery consists precisely in not constructing useless combinations, but in constructing those that are useful, which are an infinitely small minority. Discovery is discernment, selection.



A Parallel Between The Two Chapters



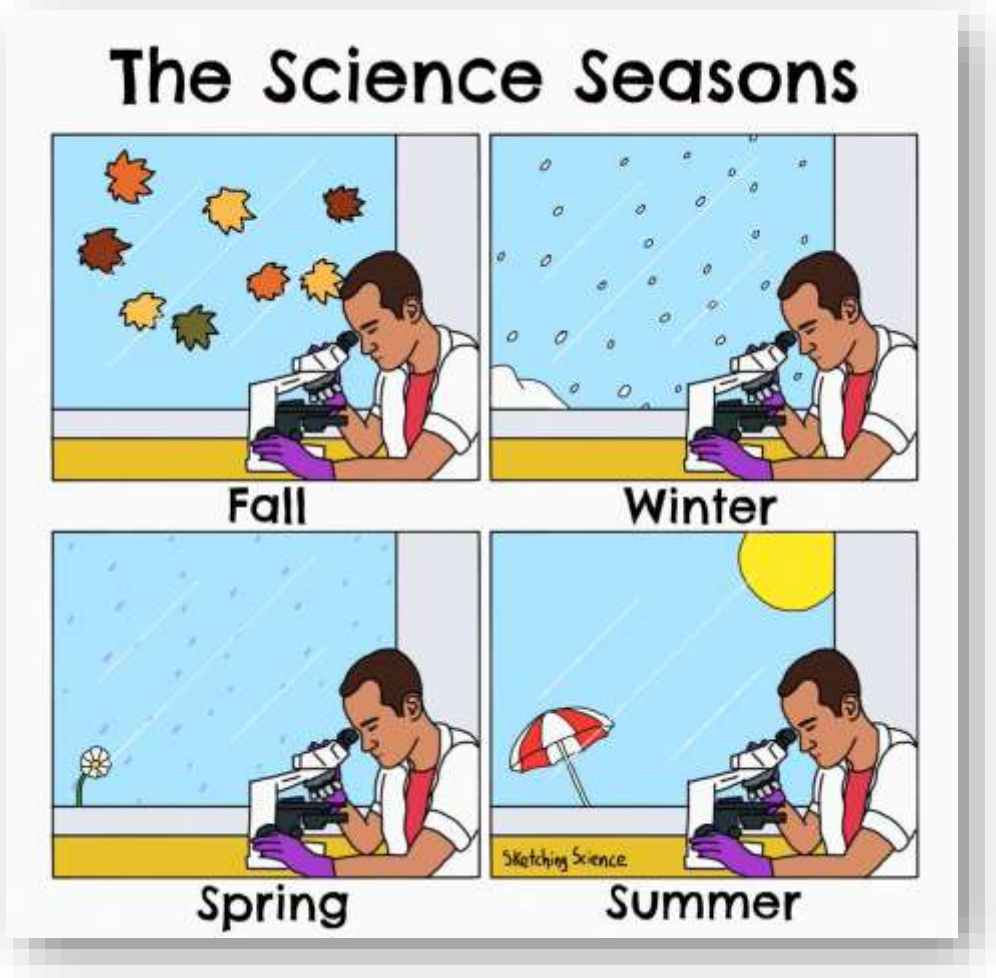
The process of mathematical discovery



Conscious working out of difficult problem

The process of scientific discovery

Conscious working out of difficult problem





The process of scientific discovery

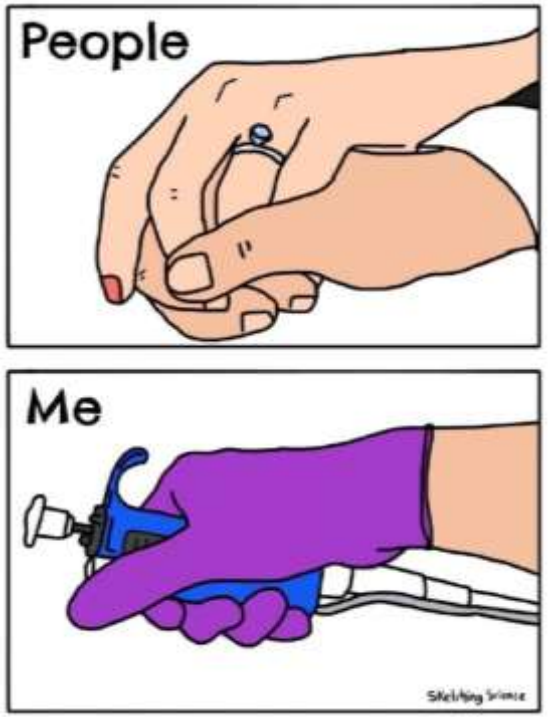
Conscious working out of difficult problem

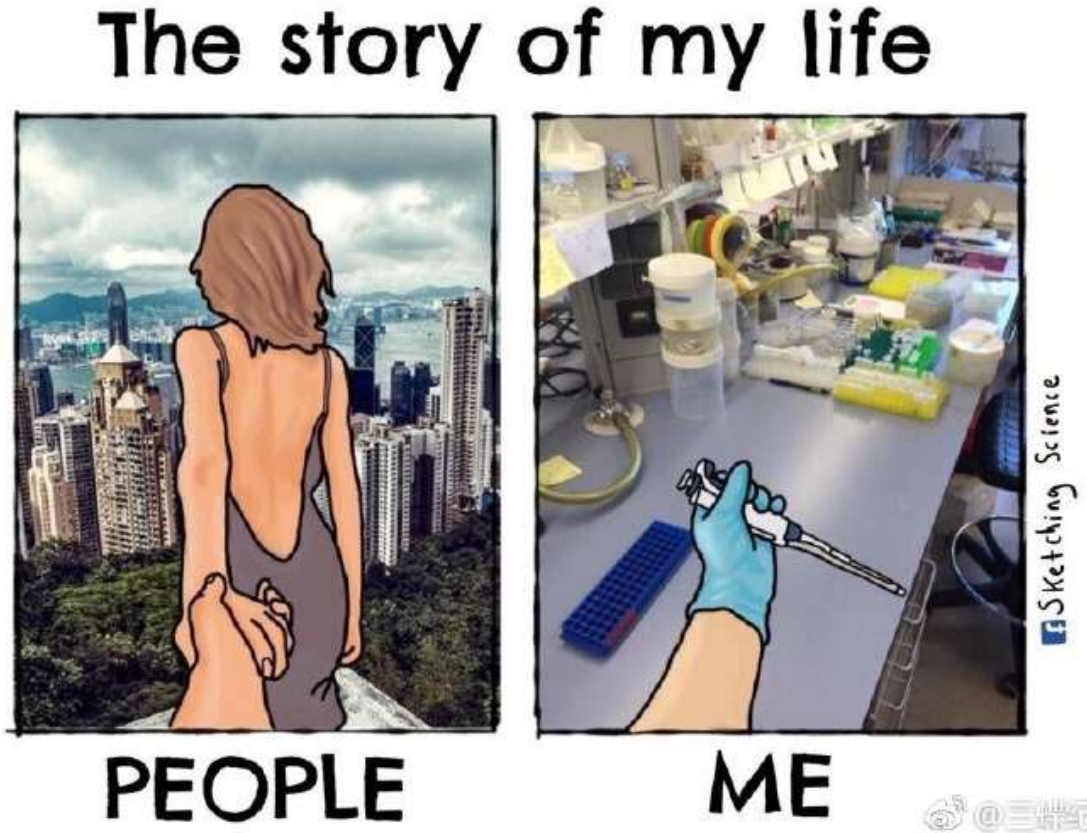


The Process of Scientific Discovery

Conscious working out of difficult problem

$$\text{秃头} = \int_{\text{入学}}^{\text{毕业}} \text{搞科研}\ dt$$

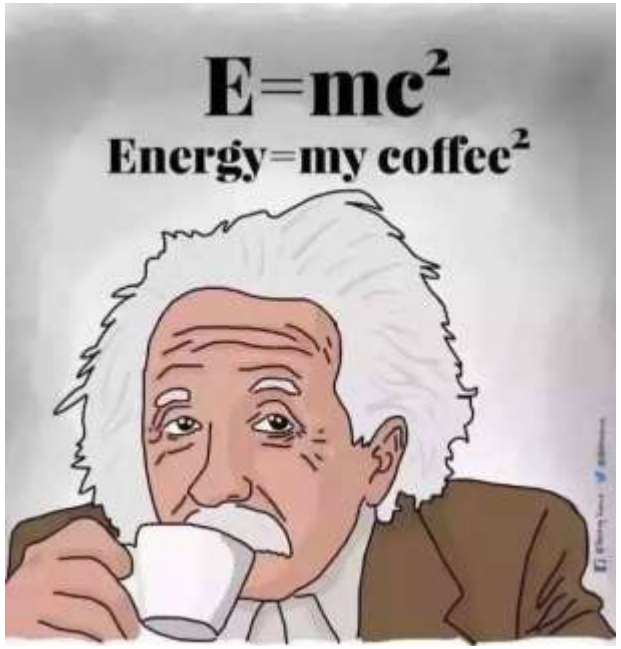




The Process of Scientific Discovery

Conscious work out of difficult problem

Rest from conscious work





The process of scientific discovery

Conscious work out of difficult problem

Rest from conscious work

Unconscious work by subliminal ego (inspiration)

Examples of Intuitive Discoveries in Literature and Science

Unconscious work by subliminal ego



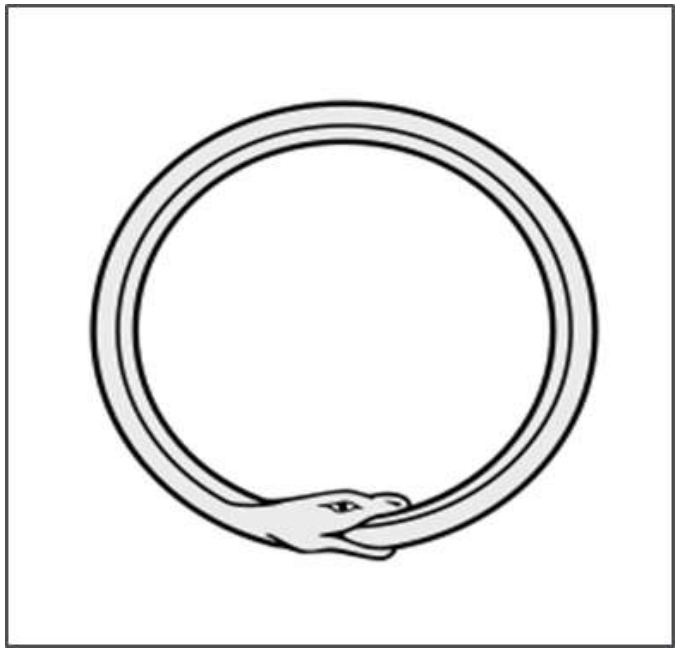
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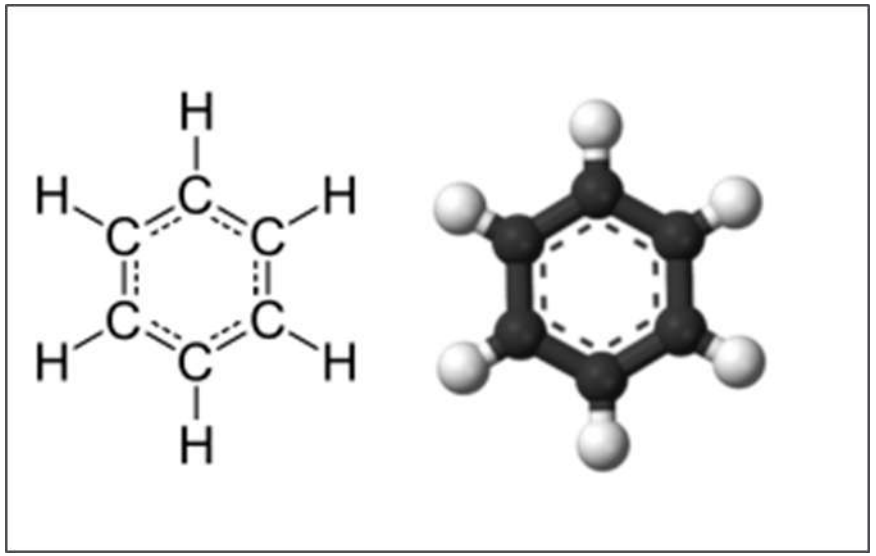
各自散后，香菱满心中还是想诗。至晚间对灯出了一回神，至三更以后上床卧下，两眼鳏鳏，直到五更方才朦胧睡去了。一时天亮，宝钗醒了，听了一听，他安稳睡了，心下想：“他翻腾了一夜，不知可作成了？这会子乏了，且别叫他。”正想着，**只听香菱从梦中笑道：“可是有了，难道这一首还不好？”**宝钗听了，又是可叹，又是可笑，连忙唤醒了他，问他：“得了什么？你这诚心都通了仙了。学不成诗，还弄出病来呢。”一面说，一面梳洗了，会同姊妹往贾母处来。**原来香菱苦志学诗，精血诚聚，日间做不出，忽于梦中得了八句。**梳洗已毕，便忙录出来，自己并不知好歹，便拿来又找黛玉。刚到沁芳亭，只见李纨与众姊妹方从王夫人处回来，宝钗正告诉他们说他梦中作诗说梦话。众人正笑，抬头见他来了，便都争着要诗看。

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–《红楼梦》第四十八回“慕雅女雅集苦吟诗”

Unconscious work by subliminal ego



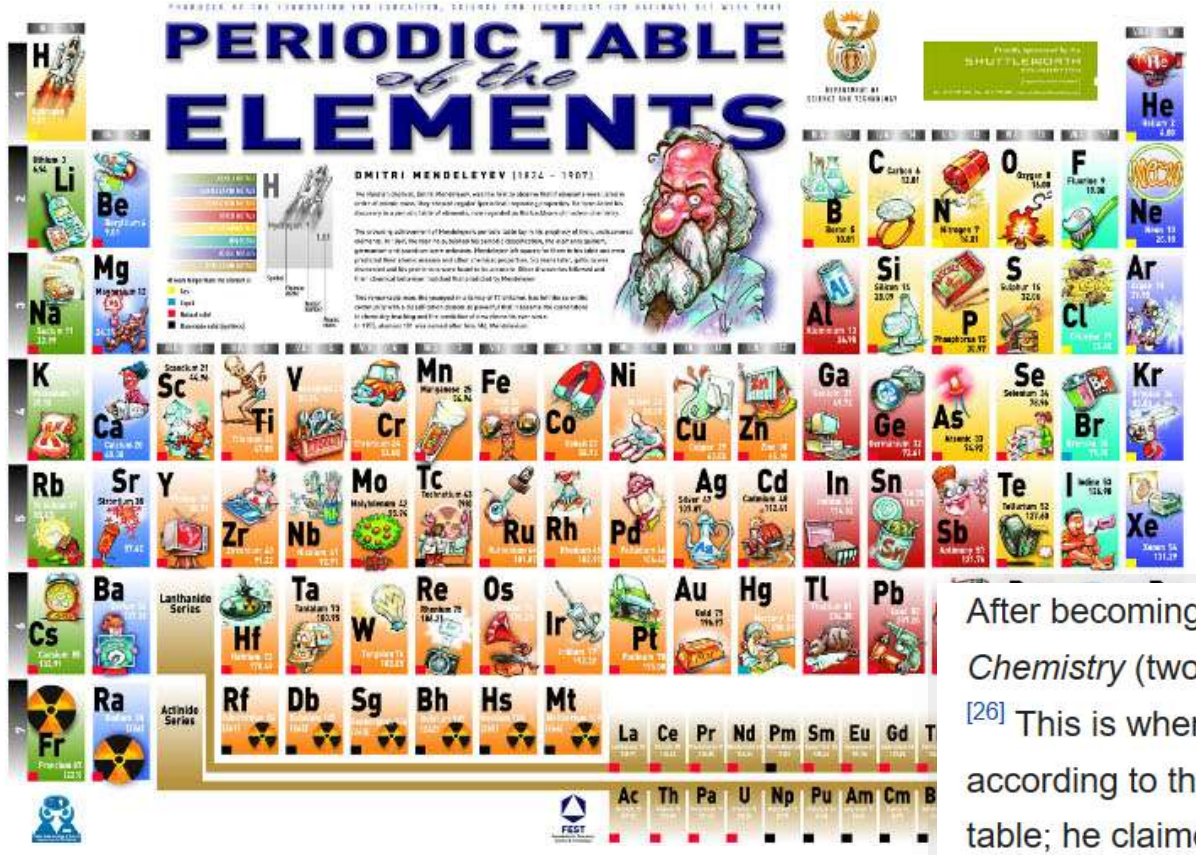


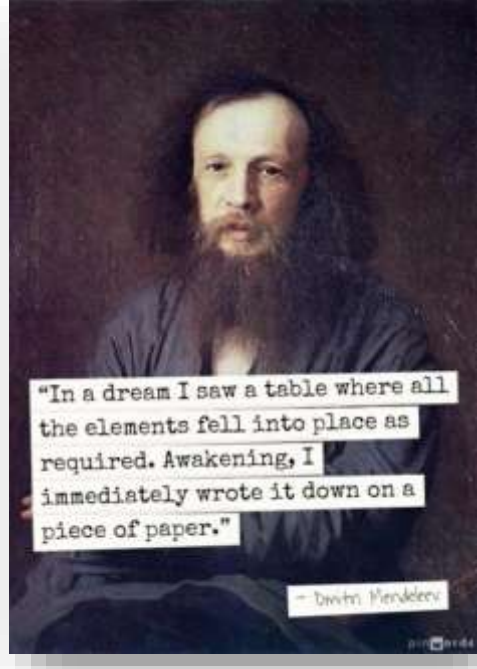


August Kekulé
(1829 –1896)

The new understanding of benzene, and hence of all aromatic compounds, proved to be so important for both pure and applied chemistry after 1865 that in 1890 the German Chemical Society organized an elaborate appreciation in Kekulé's honor, celebrating the twenty-fifth anniversary of his first benzene paper. Here Kekulé spoke of the creation of the theory. He said that he had discovered the ring shape of the benzene molecule after having a reverie or day-dream of a snake seizing its own tail (this is an ancient symbol known as the ouroboros).[11] This vision, he said, came to him after years of studying the nature of carbon-carbon bonds.

Unconscious work by subliminal ego





Dmitri Mendeleev
(1834 –1907)

After becoming a teacher in 1867, Mendeleev wrote the definitive textbook of his time: *Principles of Chemistry* (two volumes, 1868–1870). It was written as he was preparing a textbook for his course.[26] This is when he made his most important discovery.[26] As he attempted to classify the elements according to their chemical properties, he noticed patterns that led him to postulate his periodic table; he claimed to have envisioned the complete arrangement of the elements in a dream:[33][34][35][36][37]

> "I saw in a dream a table where all elements fell into place as required. Awakening, I immediately wrote it down on a piece of paper, only in one place did a correction later seem necessary."
>
> — Mendeleev, as quoted by Inostrantzev[38][39]

The process of mathematical discovery

Conscious work out of difficult problem

Rest from conscious work

Unconscious work by subliminal ego (inspiration)

Conscious work in ordering, demonstrating, and verification of the results of insight

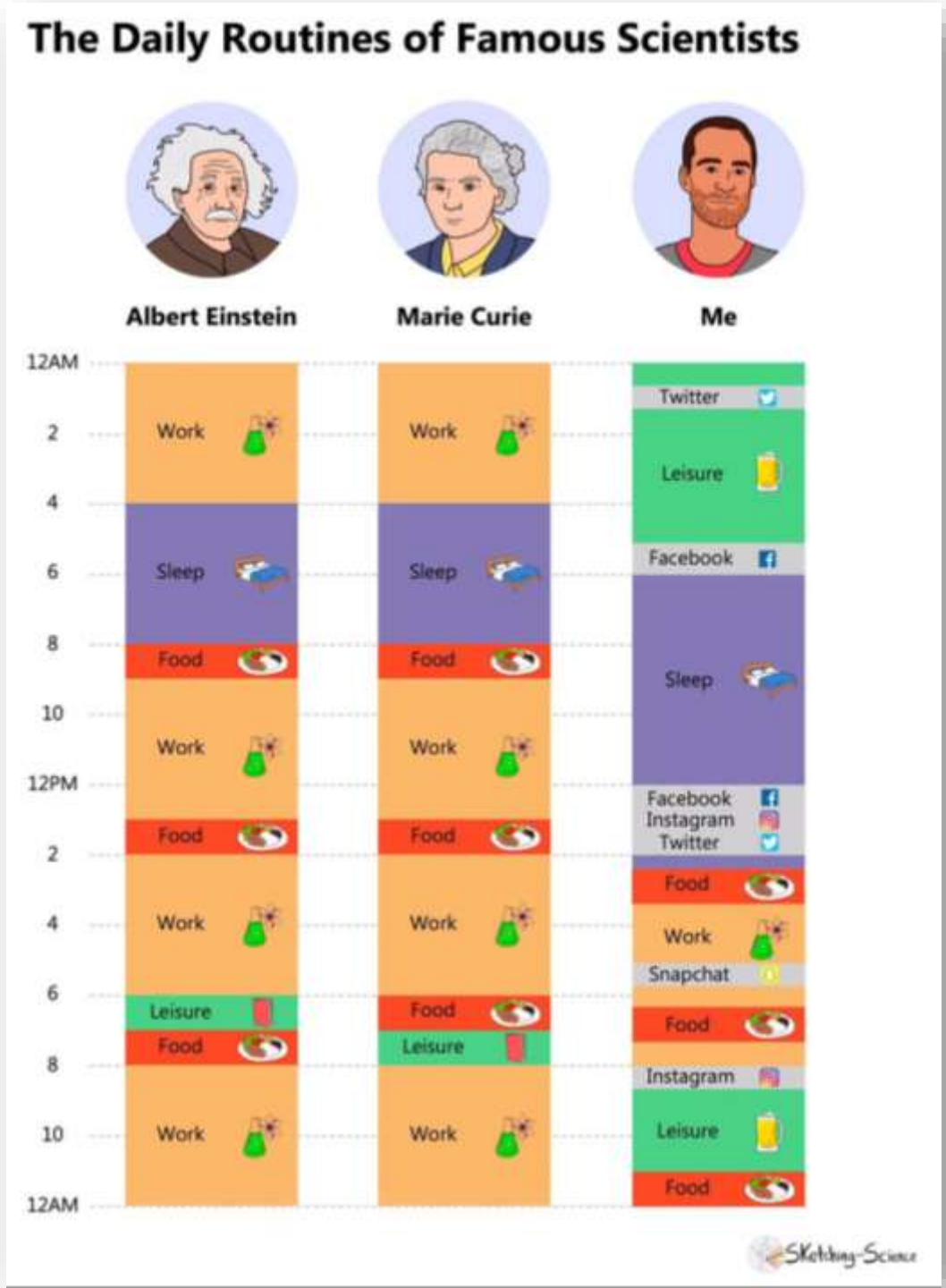
The process of mathematical discovery

Conscious working out of difficult problem

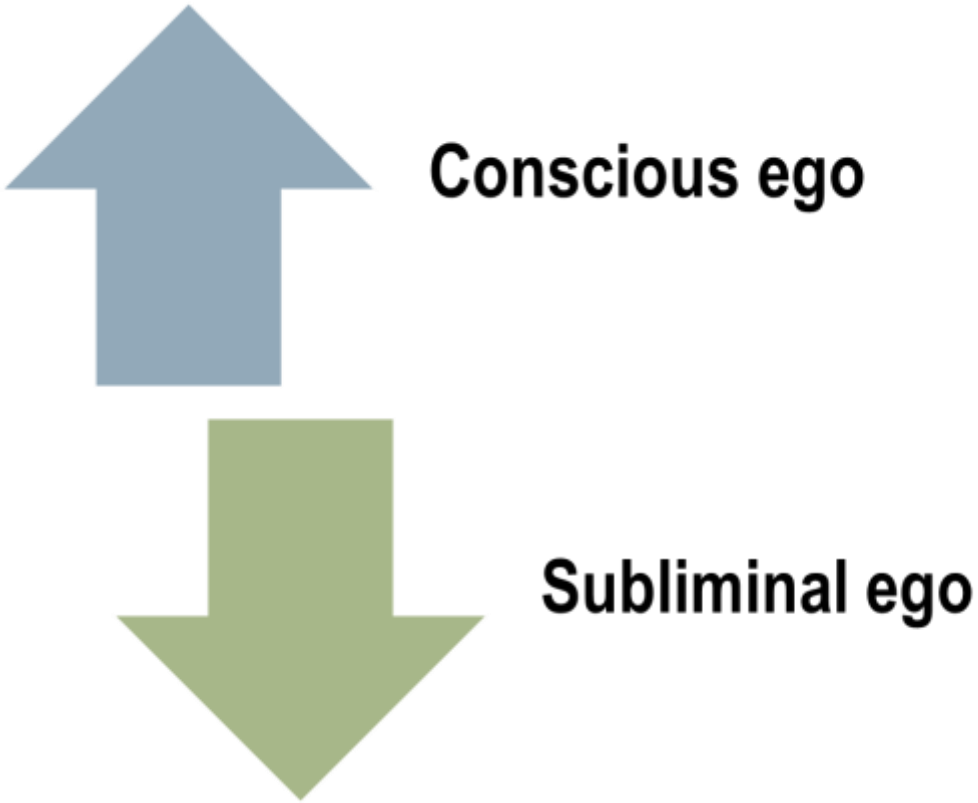
Rest from conscious work

Unconscious work by subliminal ego (inspiration)

Conscious work ordering, demonstrating, and verification of the results of insight



The process of mathematical discovery



Conscious working out of difficult problem

Rest from conscious work

Unconscious work by subliminal ego (inspiration)

Conscious work ordering, demonstrating, and verification of the results of insight

The process of mathematical discovery

古今之成大事业、大学问者，必经过三种之境界：

“昨夜西风凋碧树。独上高楼，望尽天涯路”，此第一境界也。

“衣带渐宽终不悔，为伊消得人憔悴”，此第二境界也。

“众里寻他千百度。蓦然回首，那人却在灯火阑珊处”，此第三境界也。

《人间词话》 王国维



Reading Assignment:

Finish reading Text 7.